

Program : Diploma in Renewable Energy	
Course Code : 2291	Course Title: Basic Electrical Engineering & Renewable Technology
Semester : 2	Credits: 3
Course Category: Engineering Science	
Periods per week: 3 (L:3 T:0 P:0)	Periods per semester: 45

Course Objectives:

- To impart a basic knowledge of electrostatic field and capacitance
- To introduce fundamental concepts of electromagnetic induction
- To know the basics of batteries and energy storage techniques.
- To understand the significance of renewable energy sources.

Course Prerequisites:

Topic	Course Code	Course name	Semester
Basic knowledge in Physics		Applied Physics- I	1
Basic knowledge in Mathematics		Mathematics- I	1

Course Outcomes:

On completion of the course, the students will be able to:-

COn	Description	Duration (Hours)	Cognitive Level
CO1	Identify the basic concepts of electrostatics and energy storage.	10	Applying
CO2	Identify the basic concepts of electromagnetism and energy storage.	12	Applying
CO3	Explain the charging methods and interconnections of batteries .	10	Understanding
CO4	Summarize various renewable energy technologies.	11	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

On completion of the course, the students will be able to:-

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Identify the basic concepts of electrostatics and energy storage		
M1.01	Apply the basic concepts of electrostatics to solve simple problems	3	Applying
M1.02	Explain the fundamental concepts of electric potential and lightning phenomena	2	Understanding
M1.03	Develop the expression for capacitance of a capacitor	2	Applying
M1.04	Identify various applications of capacitors .	3	Applying

Contents:

Electrostatics: Basic concepts of static electricity - Laws of electrostatics -Absolute and Relative Permittivity of a medium - Dielectric constant- Electric field- Electrostatic field - Electric flux,Electric field strength or field Intensity ,Electric flux density- Simple problems.

Electrical Potential: Potential and potential difference- Derivation of expression for potential at a point in air and in other medium- - Potential gradient- Breakdown voltage-Dielectric strength- Lightning phenomena

Storage of electric charge -Capacitor-Capacitance - Charging and Discharging - Parallel plate capacitor- Derivation of equation for capacitance - Classification of capacitors- Fixed and Variable

Capacitor connections -Derivation of equivalent capacitance in series, parallel and series-parallel -Derivation of energy stored in a capacitor -Simple problems-List the applications of capacitors.			
CO2	Identify the basic concepts of electromagnetism and energy storage		
M2.01	Explain the basic concepts of magnetism	3	Understanding
M2.02	Identify the comparison between magnetic and electric circuits	2	Applying
M2.03	Apply the laws of electromagnetic induction to solve simple problems.	5	Applying
M2.04	Identify various applications of electromagnets	2	Applying
	Series Test – I	1	
Contents: Magnetism- Coulomb's laws -Magnetic permeability-Absolute and Relative -Magnetic flux,Field intensity, Reluctance,Magneto Motive Force -Relation between MMF, flux and reluctance-Magnetization curve -Hysteresis -Hysteresis loss -B-H curve Magnetic circuit- Basic concepts of series and parallel magnetic circuits - Calculation of ampere-turns-Simple problems - Comparison between magnetic circuit and electric circuit Electromagnetism- Force on a current carrying conductor lying in a magnetic field (equation only)- Fleming's Left Hand Rule Electromagnetic Induction:- Faraday's Laws of Electromagnetic Induction-Direction of induced emf and currents- Lenz's law- Fleming's Right Hand Rule, Statically and dynamically induced emf-Self induction and mutual induction -Self inductance and Mutual inductance-Coefficient of coupling (definition and equation only) -Simple problems Energy stored in a magnetic field- Derivation of Energy stored in a magnetic field (or inductor)-Lifting power of electromagnet -Applications of electromagnets			
CO3	Explain the charging methods and interconnections of batteries .		
M3.01	Classify various energy forms and storage methods.	2	Understanding
M3.02	Identify the classification and applications of batteries	4	Understanding
M3.03	Summarize the charging methods and maintenance of batteries.	2	Understanding
M3.04	Illustrate the methods of interconnection of batteries.	2	Understanding

Contents :

Energy Storage- Various forms and methods - Mechanical-Electrical-Chemical - Thermal - Biological.

Batteries-Introduction- Definitions- Ampere hour rating- Ampere hour efficiency,- Battery backup- Depth of discharge.Classification (Listing only)

Lead Acid Batteries -Principle of operation -Construction-SLI-Tubular-Sealed portable (Introduction and properties only)

Nickel Cadmium Batteries - Principle of operation - Construction - Advantages - Disadvantages. Basic introduction of Lithium Ion Batteries -Applications.

Charging Methods- Basic concepts of charging methods - -Specifications-Maintenance-Use of hydrometers.

Interconnection of batteries-Importance -Types of connections -Series ,Parallel ,Series-Parallel .

CO4	Summarize various renewable energy technologies		
M4.01	Classify various energy sources	1	Understanding
M4.02	Explain the significance of renewable energy sources.	1	Understanding
M4.03	Outline the Basic concepts of Renewable Technologies	6	Understanding
M4.04	Summarize the concepts of additional alternate energy sources	3	Understanding
	Series Test – II	1	

Contents :

Sources of energy - Classifications - Primary and Secondary-Commercial and Non commercial - Conventional and Non conventional Sources . (Definitions and examples only)

Renewable energy sources:Scope , need and advantages- Environmental impacts - Greenhouse effect and Global warming

Renewable Technologies - Basic concepts and schematic diagram of power generation from Solar energy,Wind energy and Ocean energy-Basic concepts of Biomass , Biogas ,Geothermal energy, Chemical energy Sources and Hydrogen energy.

Additional alternate energy sources-MHD power generation , Thermo Electric Power,Thermionic Generation, Thermo Nuclear Fusion Energy(Basic concepts only)

Text / Reference

T/R	Book Title/Author
T1	A Textbook of Electrical Technology Vol I-B L Theraja, A.K Theraja-S.Chand & Company Ltd.
T2	Non Conventional Energy sources- G D Rai-Khanna Publishers
R1	Bhattacharya, S. K., Basic Electrical & Electronics Engineering, Pearson
R2	D P Kothari and I J Nagrath, “Basic Electrical Engineering”, Tata McGraw Hill, 2010.
R3	D C Kulshreshtha, “Basic Electrical Engineering”, Tata McGraw Hill, 2010.
R4	Non-conventional Energy Resources-B H Khan Mc Graw Hill Education
R5	D.P. Kothari, K. C. Singal, Rakesh Ranjan , “Renewable Energy Sources and Emerging Technologies” Prentice Hall India Learning Private Limited

Online Resources

Sl.No	Website Link
1	https://circuitdigest.com/article/different-types-of-batteries
2	https://www.physics-and-radio-electronics.com/electromagnetics/electrostatics/introduction.html
3	https://www.electrical4u.com/
4	https://nptel.ac.in/courses/121/106/121106014/ : course name Non conventional Energy Resources by IIT Madras
5	https://nptel.ac.in/content/storage2/courses/108103009/download/M9.pdf